Provenance first: how far do three fatigue endpoints carry a new condition?

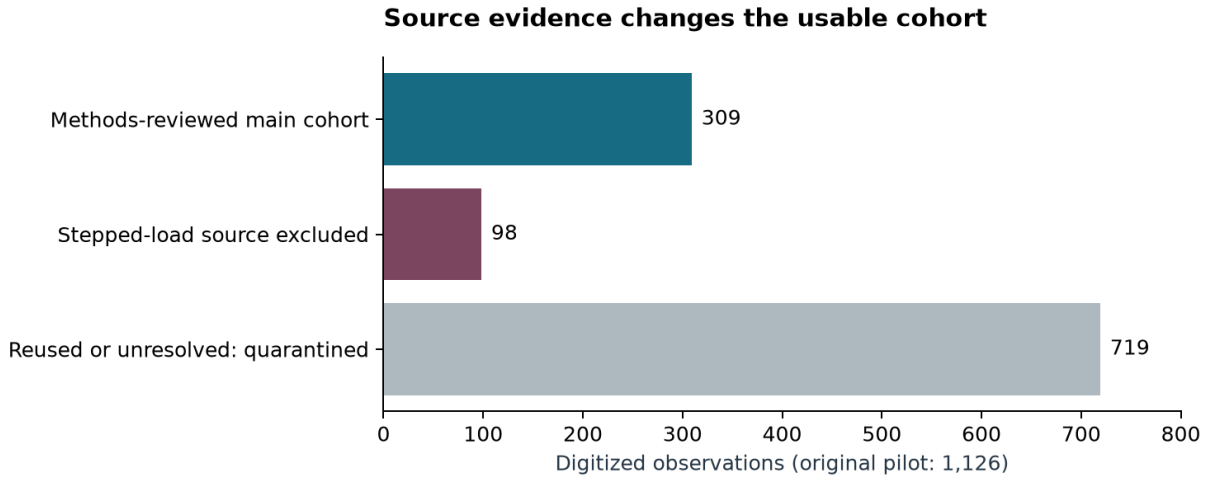
A source-audited, censored AM316L benchmark with an acquired external campaign. The prespecified useful-calibration criterion was not met.

Finding

In a retrospective full-training model-based track, three endpoints per new condition increased the geometric lower-life limit by 1.92x while the source-balanced danger identification interval fell from 13.35-15.41% to 8.31-8.93%. The descriptive source-bootstrap gain interval is 0.84-3.65x and includes no improvement. Some sources became less safe. This is a promising aggregate trend, not a demonstrated reliable-design algorithm.

Why provenance had to precede fitting

The original public-data pilot contained 1,126 approximately digitized points. Original-methods review identified 98 records from Merot et al. as stepped-load/damage-equivalent observations rather than ordinary independent constant-amplitude lifetimes. They were excluded. Further publication-level reuse, unresolved figure-to-experiment mappings and a maximum-stress/amplitude conflict required quarantine. The main cohort contains 309 points, 32 database curves and 11 source blocks with reviewed experimental methods. These are not 11 certified independent laboratories, nor are database row IDs physical specimen IDs.



Methods reviewed does not certify every point-to-specimen identity. Raw data are unchanged.

Additional source-supported corrections: one Wood endpoint labelled runout was a fracture; one Kedziora Renishaw group inherited an unsupported heat treatment. Corrections are applied only to derived phase-2 data. Raw files and v1 outputs remain immutable. These findings concern our open-data benchmark and do not prove an error in Centola et al.'s unavailable 518-row dataset. [1-4]

Frozen question and models

For an unseen provenance source, do 3, 5 or 10 endpoints from one new condition improve useful lower-life limits? All other curves from that source remain outside population fitting and calibration. Conditions mean source S-N series, not verified build batches. All model features and population parameters are learned from historical fitting sources.

Panel	Endpoints per condition	Eligible curves / sources	Fixed-query minimum
A: primary	0, 3	18 / 9	5
B: secondary	0, 3, 5	14 / 6	5
C: limited secondary	0, 3, 5, 10	6 / 5	5

Seeds 0, 17 and 42 define outcome-blind record-ID permutations. Reserve the maximum support pool once per panel and use all remaining records as the same queries for every k. The panels have different populations; they are not a single learning curve. Runouts contribute a survival likelihood, not an exact failure density.

Baselines: (i) pooled monotone censored lognormal Basquin; (ii) a target-only intercept update with historical slope/scatter; (iii) a condition-random-intercept Basquin model, updating a proper target intercept posterior with censored support. The predictive lower decile integrates specimen scatter and target intercept uncertainty. Population parameters are empirical Bayes estimates; the main bounds omit their estimation uncertainty. A training-source bootstrap mixture with 20 replicates from 11 historical sources assesses this externally. A continuous two-slope monotone curve with a training-only median stress knot is a prespecified curvature sensitivity.

Calibration and claim discipline

The original primary criterion required a useful 90% endpoint-calibrated bound, risk at most 10%, a paired risk-increase limit and a greater-than-1.25x life gain supported by descriptive source-bootstrap intervals. Source review leaves only 4-6 calibration blocks when at least four fitting blocks are retained. The required finite-sample rank exceeds the calibration count, so the literal bound is zero. It is safe but useless and fails the utility criterion. This finite-rank result applies to this frozen calibration protocol, not to every parametric or survival method.

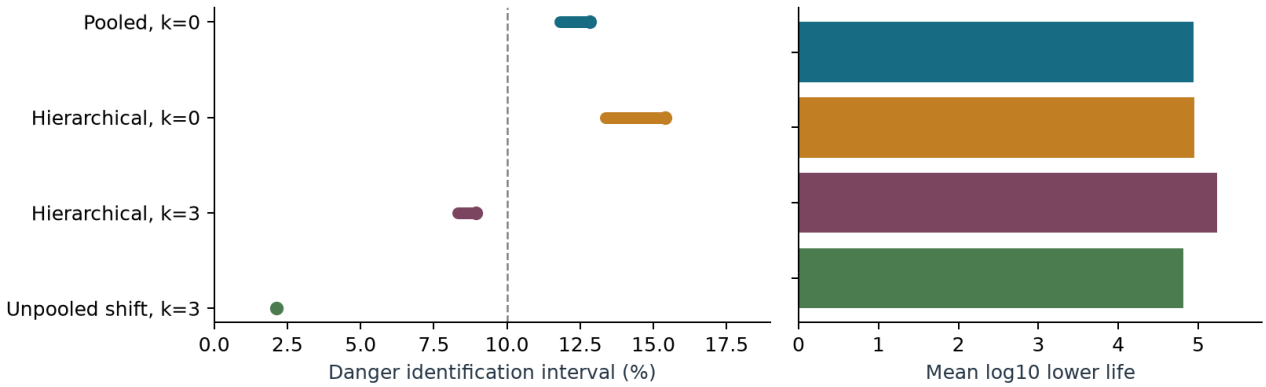
A separately recorded full-training raw track fits all other verified sources and reports model-based 10th percentiles. It does not replace the failed calibrated primary test. Hierarchical borrowing and modern survival conformal prediction are established prior work. Missing potential censoring times on failures rule out direct Type-I calibration, but modern right-censoring methods exist; none guarantees arbitrary manufacturing-source shift. [5-8]

Risk and utility

Danger is L>T. For a runout C<T, L<=C establishes safety for that event; L>C remains ambiguous. Report [known violations/n, (known+ambiguous)/n]. These are censoring identification endpoints, not confidence intervals. Average within curve, then source, then seeds. Utility is paired log10(Lk/L0) on identical queries; larger unsafe limits are not useful. Source bootstrap intervals are descriptive because outer models reuse training sources.

Internal results: average gain is not source reliability

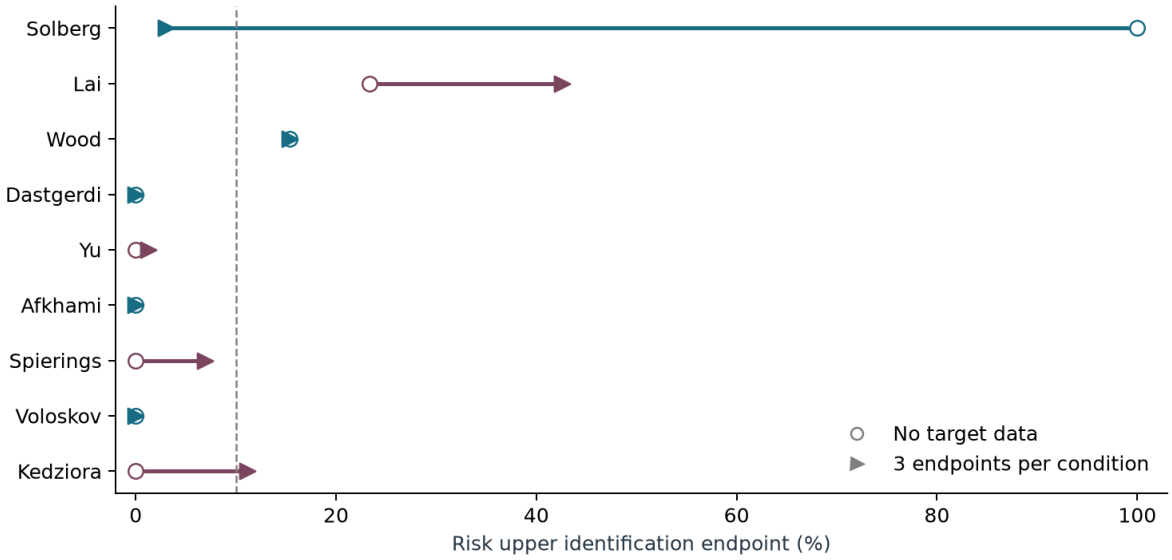
Three endpoints per condition: useful average, uncertain generalization



Panel A: 18 curves / 9 target sources, full historical training. Intervals describe censoring, not confidence.
Unpooled shift availability: 98.15%; other methods: 100%. No method superiority is established.

The full-training raw A-panel gain is 1.92x (0.84-3.65x descriptive 95% interval). The upper bootstrap endpoint for the upper identified true-risk increase is +9.30 percentage points, above the 2-point noninferiority margin. Hence even the secondary raw result does not meet the intended robust positive-claim standard.

The mean hides source-specific risk increases



Source-balanced descriptive estimates, seed averages. These are not individual-source safety certificates.

Solberg's source risk upper endpoint falls from 100% to 3.03%, driving most of the mean improvement. Lai increases from 23.33% to 42.60%; Spierings from 0% to 6.94%; Kedziora from 0% to 11.11%. These are small-source descriptive estimates, not certificates. Deleting the unfavorable sources or treating seed repeats as new experiments would be invalid.

Acquired external campaign: the limits become shorter and less dangerous

Trombley, Birnbaum and Allison’s public Materials Commons pages provide 129 complete named as-built specimens: 44/40/45 at 1.5/2.5/5.0 mm diameter. The article states 130; one incomplete page is excluded without imputation. All 154 available fatigue records and the 129-record subset were independently checked against the public HTML. No strong historical-coordinate reuse signal was found. This is one campaign with three related conditions. Four builds are reported, but specimen-to-build mapping is not certified. [9-10]

Tests were performed in 2019-2021, at 20 kHz and R=-1. The failure endpoint is a resonance-drop/visible-crack criterion, potentially before full fracture. Thus this is a cross-campaign, cross-endpoint stress test, not clean validation of an identical fracture-time distribution. Preserve actual runout endpoints; do not invent a universal potential censoring time for failures.

Mechanism / k	Danger fraction	Geometric lower life
Random / 0	53.66-53.66%	3,646,481 cycles
Random / 3	13.19-13.19%	778,760 cycles
Random / 5	16.25-16.25%	998,370 cycles
Random / 10	15.03-15.03%	1,159,981 cycles
Chronological / 0	48.53-48.53%	3,915,926 cycles
Chronological / 3	1.96-1.96%	443,883 cycles
Chronological / 5	1.96-1.96%	568,477 cycles
Chronological / 10	3.07-3.07%	613,991 cycles

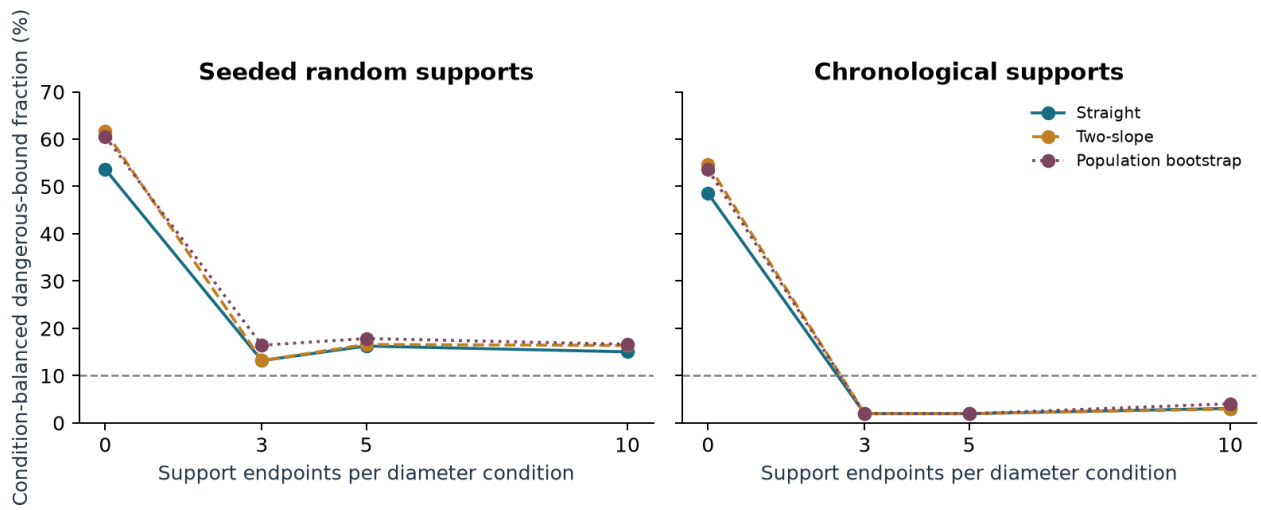
k is per diameter condition: k=3 uses nine support endpoints in total. Each mechanism and seed keeps 99 common queries over its k values. Random and chronological mechanisms have different query sets, so their difference cannot identify a causal benefit of a selection policy. One campaign provides no independent campaign-level confidence interval.

Random k3 reduces danger from 53.66% to 13.19%, still above 10%, by lowering the life limit about 4.68x rather than extending it. The 20-population-bootstrap mixture has 16.42% random-k3 danger; the two-slope model has 13.20%. Chronological k3 gives 1.96% on its own query population. We retain every result rather than select the most favorable mechanism.

The original article uses a Weibull random-fatigue-limit model. Our finite curvature control is not its reproduction, and superiority over that relevant physical comparator is untested. This limitation must remain in any communication. [9,11]

Controls, sensitivities and verification

An acquired external campaign: adaptation helps, reliability remains conditional



129 named specimens, 3 diameter conditions, ONE campaign; 99 fixed queries per mechanism/seed.
k=3 uses 9 supports in total. Query sets differ between mechanisms: their difference is not a causal selection effect.

An explicitly post-result diagnostic holds the hierarchical population slope/scatter fixed and replaces prior borrowing with an unpooled censored target intercept. At k0 its reference equals the same hierarchical predictive distribution; all-runout supports remain unavailable. This control avoids attributing a changed population scatter to a benefit from shrinkage. It was not used to rescue the original primary claim.

On matched internal k3 queries, hierarchical versus this simple control gives about 1.10x life, with a descriptive interval including no gain. Externally on random supports, the control has 6.99% danger versus 13.19% for the hierarchy, at approximately 1/1.42 of its life limit. There is a tradeoff, not demonstrated prior-specific superiority. Availability and common-query masks are retained.

Near-coordinate clustering removes one historical point; the A-panel risk endpoints remain unchanged. Excluding Afkhami, whose stress-axis visual verification remains unresolved, gives k3 risk 9.00-9.70%. Full-family/weak-edge and broader database-labelled analyses are sensitivity results, not replacements for reviewed-source evidence. They retain unresolved original-experiment and stress-definition issues.

All 31 synthetic model/evaluation tests pass. Two independently written arithmetic checks reproduce core source-balanced summaries and paired censoring contrasts. Direct scalar numerical integration checks mixed-censor posteriors. Critical strict, full-training and external predictions were replayed byte for byte. These establish computational consistency, not independent physical replication. Fit statuses, split IDs, hashes and exact commands are saved.

Boundaries

The cohort remains digitized with incomplete physical specimen/batch IDs. Several method-reviewed sources lack exhaustive point matching. Original author endpoints and sampling protocols differ; staircase-selected stresses and retrospective random supports are not an optimized sequential test policy. Counts of endpoints are not measured machine hours or monetary savings. We did not run new physical experiments, obtain Centola’s original data/code, reproduce the BNN/Owen calculations, or establish general source-conditional coverage.

Contribution assessment and the next discriminating experiment

The defensible contribution is a provenance-corrected benchmark, concrete source/label corrections and a documented counterexample to interpreting average few-shot improvement as reliable performance in every new condition. The hierarchical method itself is established. This package supports a focused technical discussion with Tridello, not a claim of a new validated fatigue-design algorithm.

The next question should be: with a fixed test budget, which stress levels reveal a harmful new-condition shift and produce the largest useful lower-life limit at acceptable source-specific risk? First secure physical specimen/build IDs, consistent failure definitions and actual censoring plans. Then benchmark the published random-fatigue-limit model and the matched simple intercept control. Freeze one common query set across random, stress-spanning and chronological support policies; choose any policy only using development campaigns. Test on additional independent campaigns and report source-specific failures, not only an average. A new campaign and a frozen protocol are required for a strong confirmatory claim.

References and evidence

[1] Zhang & Xu. FatigueData-AM2022 v2. Data DOI 10.6084/m9.figshare.22337629.v2; descriptor DOI 10.1038/s41597-023-02150-x. CC BY 4.0.

[2] Merot et al. Observations on process/corrosion defects. DOI 10.1016/j.ijfatigue.2021.106552. Original methods section 2.4/Table 3; source-method evidence in provenance_report.md.

[3] Wood et al. Materials 12,4203 (2019). DOI 10.3390/ma12244203. Sections 2.1/2.9/3.4.1, Figures 9/14. https://pmc.ncbi.nlm.nih.gov/articles/PMC6947212/

[4] Kedziora et al. Materials 15,6278 (2022). DOI 10.3390/ma15186278. Sections 2/3.3. https://pmc.ncbi.nlm.nih.gov/articles/PMC9505810/

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[11] Pascual & Meeker (1999). Estimating fatigue curves with the random fatigue-limit model. DOI 10.1080/00401706.1999.10485925.

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